CNC Procedure 2015_009

Sep 1st

2015

Operation History Decoding Procedure

[Applicable for FS16/18/21/0 – i Series]

Inputting and Outputting the Operation History Data:

Recorded data can be output to an input/output unit connected via a reader/punch interface.
An output record can be input from the input/output unit.
Set the input/output unit to be used in setting parameters No. 0020 and 0100 to 0135.
To output the data, set a code in the ISO bit of a setting parameter (bit 1 of parameter No. 0020).

Output
(1) Select EDIT mode.
(2) Press the SYSTEM key, then select the operation history display screen.
(3) Press the soft keys **[(OPRT)]**, **[PUNCH]**, and **[EXEC]** in this order.
The data output to the FANUC Floppy Cassette or FANUC FA Card is stored under file name OPERATION HISTORY.

Input
(1) Select EDIT mode.
(2) Press the SYSTEM key, then select the operation history display screen.
(3) Press the soft keys **[(OPRT)]**, **[READ]**, and **[EXEC]** in this order.
1. MDI/soft key
2. Signal
3. Alarm
4. For extension (date or time)
5. MDI/soft key of path 2
6. Signal of path 2
7. Alarm of path 2

Output data format
The header and recorded operation data are output, in this order.
The operation history data is divided into four parts by identifier words.
Data other than the identifier words depends on the type.
T0 : Header
T50 : MDI/soft key
T51 : Signal
T52 : Alarm
T53 : For extension (date or time)
T54 : MDI/soft key of path 2
T55 : Signal of path 2
T56 : Alarm of path 2

1) Header

T	0	C	O	P	E	R	A	T	I	O	N	

	H	I	S	T	O	R	Y	;

C: Data word

2) MDI/soft key

T	5	0	P	0 to 1	H	*	*	;

P0: Usually
P1: At power–on
H **: Key code (See the following table.)

3) Signal

T	5	1	P	0 to 6	N	0 to 255	H	*	*	,	*	*	;	

New data Old data

P0: X0000 and above
P2: G0000 and above
P4: Y0000 and above
P6: F0000 and above
N***: DI/DO number
H **: Signal information data (hexadecimal)

4) Alarm

T	5	2	P	0 to 10	N	*	*	*	*	;	

P0: P/S No. 100
P1: P/S No. 000
P2: P/S No. 101
P3: P/S No. 0001 to 254
P4: Overtravel alarm
P5: Overheat alarm
P6: Servo alarm
P7: System alarm
P8: APC alarm
P9: Spindle alarm
P10: P/S alarm No. 5000 to 5999
P15: External alarm
N****: Alarm number (for P/S alarm, system alarm, and external alarm only)

5) For extension (date or time)

Date

T	5	3	P	0 to 1	E	0	D	*	*	*	*	*	*	*	*	;

Time

T	5	3	P	0 to 1	E	1	D	*	*	*	*	*	*	;

P0: Usually
P1: At power–on
E0: Date
E1: Time
D*..*: Data Example) June 29, 2001

D	2	0	0	1	0	6	2	9

6) MDI/soft key of path 2

T	5	4	P	0 to 1	H	*	*	;	

P0: Usually
P1: At power–on
H **: KCB code (See the following table.)

7) Signal of path 2

T	5	5	P	2, 6	N	1000 to 1255		H	*	*	.	*	*	;	
									New data		Old data				

P2: G1000 and above
P6: F1000 and above
N***: DI/DO number
H **: Signal information data (hexadecimal)

8) Alarm of path 2

T	5	6	P	0 to 10	N	*	*	*	*	;	

P0: P/S No. 100
P1: P/S No. 000
P2: P/S No. 101
P3: P/S No. 001 to 254
P4: Overtravel alarm
P5: Overheat alarm
P6: Servo alarm
P7: System alarm
P8: APC alarm
P9: Spindle alarm
P10: P/S alarm No. 5000 to 5999
P15: External alarm
N****: Alarm number (for P/S alarm, system alarm, and external alarm only)

Notes:

(1) While the operation history screen is displayed, no information can be recorded to the history.
(2) An input signal having an on/off width of up to 16msec is not recorded in the history. Some signals are not recorded in the history.
(3) Once the storage becomes full, old data is deleted, starting from the oldest record. Up to about 8000 key information items can be recorded.
(4)The recorded data is retained even after the power is turned off. A memory all clear operation, however, erases the recorded data.
(5)The operation history function cannot execute sampling when the OHS bit (bit 7 of parameter No. 3106) is set to 1.
(6) Set the date and time on the setting screen.
(7) The time needed to input and output 6000 operation records at a rate of 4800 baud is as follows:

Output: About 5 minutes
Input: About 2 minutes and 30 seconds
This file corresponds to a paper tape of about 180 m in length.

Decoding of KCB (KEY CODE BIT) – 1/2:

Key codes (MDI/soft key)
(00H to 7FH)

	0	1	2	3	4	5	6	7
0			Space	0	@	P		
1			!	1	A	Q		
2			"	2	B	R		
3			#	3	C	S		
4			$	4	D	T		
5			%	5	E	U		
6			&	6	F	V		
7			'	7	G	W		
8			(	8	H	X		
9			)	9	I	Y		
A	; (EOB)		*	:	J	Z		
B			+		K	[		
C			'	<	L	¥		
D			–	=	M	]		
E			.	>	N			
F			/	?	O	–		

Decoding of KCB (KEY CODE BIT) – 2/2:

(80H to FFH)

	8	9	A	B	C	D	E	F
0		Reset *						F0 *
1		MMC *						F1 *
2		CNC *						F2 *
3								F3 *
4	Shift	Insert *						F4 *
5		Delete *						F5 *
6	CAN	Alter *						F6 *
7								F7 *
8	Cur→ *	Input *					POS *	F8 *
9	Cur← *						PROG *	F9 *
A	Cur↓ *	Help *					OFFSET SETTING *	
B	Cur↑ *						SYSTEM *	
C							MESSAGE *	
D							CUSTOM GRAPH "1"	
E	Page↓ *						CUSTOM *	FR *
F	Page↑ *						Fapt *	FL *

*1:On the small–sized keypad, ED corresponds to the [CUSTOM GRAPH] key. On a standard keyboard, ED corresponds to the [GRAPH] key and EE to the [CUSTOM] key.

* : Command key

For Example the sample is as below:

Down loaded Data	Decoded Data
T0COPERATION HISTORY	T0COPERATION HISTORY
T53P1E1D190332	At Power On # 190332
T52P6	Servo Alarm
T53P0E0D20150727	Date # 20150727
T53P0E1D190343	Time#190343
T50P0HEC	Message Key
T53P0E1D191200	Time#191200
T53P0E0D20150727	Date # 20150727
T53P0E1D191252	Time#191252
T53P1E0D20150727	At Power On # 20150727
T53P1E1D191340	At Power On # 191340
T52P15N1465	External Alarm # N1465
T53P0E0D20150727	Date # 20150727
T53P0E1D191411	Time#191411
T52P6	Servo Alarm
T53P0E0D20150727	Date # 20150727
T53P0E1D191443	Time#191443
T50P0HEC	Message Key
T53P0E1D192200	Time#192200
T50P0HE8	Position Key
T53P0E0D20150727	Date # 20150727
T53P0E1D192423	Time#192423
T53P1E0D20150727	At Power On # 20150727
T53P1E1D192503	At Power On # 192503
T52P6	Servo Alarm
T53P0E0D20150727	Date # 20150727
T53P0E1D192540	Time#192540
T53P0E0D20150727	Date # 20150727
T53P0E1D192607	Time#192607
T53P1E0D20150727	At Power On # 20150727
T53P1E1D192645	At Power On # 192645
T52P6	Servo Alarm
T53P0E0D20150727	Date # 20150727
T53P0E1D192745	Time#192745
T53P0E0D20150727	Date # 20150727
T53P0E1D192813	Time#192813
T53P1E0D20150727	At Power On # 20150727
T53P1E1D193449	At Power On # 193449
T52P6	Servo Alarm
T53P0E0D20150727	Date # 20150727
T53P0E1D193546	Time#193546
T50P0HEC	Message Key
T53P0E0D20150727	Date # 20150727
T53P0E1D193758	Time#193758
%	%